Texas A&M University
College of Liberal Arts
Department of Economics
Intermediate Macroeconomics, ECON 410
Spring 2026
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Final Exam

May 1, 2026

On my honor, as an Aggie, I have neither given nor received unauthorized aid on this academic work

Signed: ______________________________

NAME: ______________________________

34 questions in total. All questions have equal weight.
You have 120 minutes to complete the exam. Good luck!

Some useful formulas:

- Baseline Solow model (and also baseline endogenous growth model): law of motion for capital per worker: $\Delta k_t = \sigma f(k_t) - \delta k_t$
- Golden Rule (baseline Solow, $n = 0$ and $g = 0$): $MPK = \delta$
- Keynesian Cross government spending multiplier: $\dfrac{1}{1 - MPC}$
- Keynesian Cross tax multiplier: $\dfrac{-MPC}{1 - MPC}$
- Fisher equation: $i = r + \pi^e$

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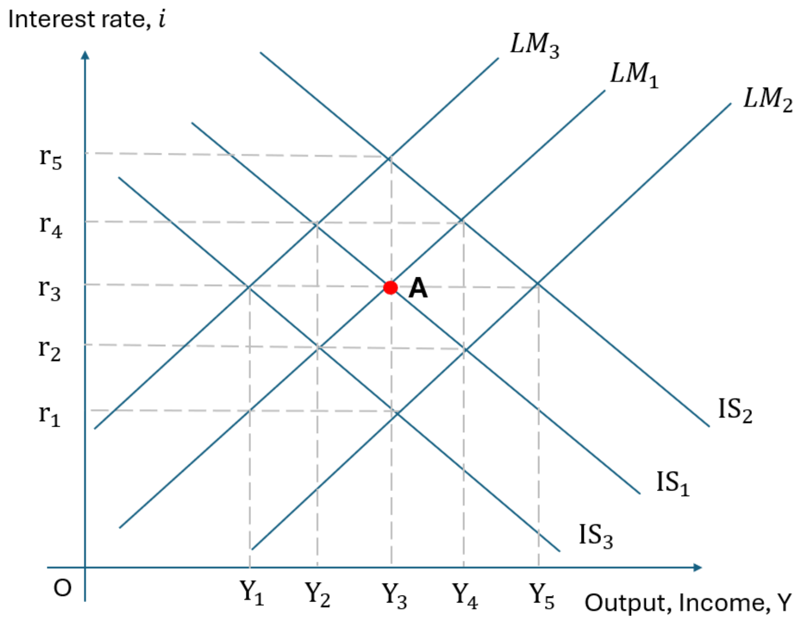
1. Which of the following options is considered as a supply shock?
   (a) All of the below
   (b) A pandemic disrupts supply chains and forces factory closures, creating shortages of raw materials and components that drive up production costs for firms.
   (c) New regulations requiring expensive pollution controls increase manufacturing costs.
   (d) A large oil price drop, resulting in lower gas prices
   (e) A drought destroys crops, raising food prices and reducing agricultural output.

2. A downward-sloping investment function yields a downward-sloping IS curve, and a downward-sloping demand for real money balance curve yields an upward-sloping LM curve.
   (a) True
   (b) False

3. An adverse __________ shock results in __________ price level and __________ in output in the short run, and a positive __________ shock results in increase in output and no change in price level in the short run, if we assume that prices are sticky.
   (a) supply; lower; lower; demand
   (b) demand; higher; lower; supply
   (c) supply; higher; lower; demand
   (d) demand; lower; higher; supply

4. Based on the graph, starting from equilibrium at point A (combination of interest rate r_3 and income Y_3), if there is an increase in government spending, then in order to keep output constant, the Federal Reserve should ___________ the money supply, shifting to ___________.



(a) not change; LM_1

(b) increase; LM_2

(c) decrease; LM_2

(d) increase; LM_3

(e) decrease; LM_3

5. The IS curve slopes downward because a ___________ interest rate reduces ___________ and thereby income.

(a) higher, money demand

(b) lower, planned investment

(c) lower, money demand

(d) higher, planned investment

6. An increase in investment demand for any given level of income and interest rates—due, for example, to more optimistic "animal spirits"—will, within the IS–LM framework, ___________ output and ___________ interest rates.

(a) increase; lower

(b) increase; raise

(c) lower; lower

(d) lower; raise

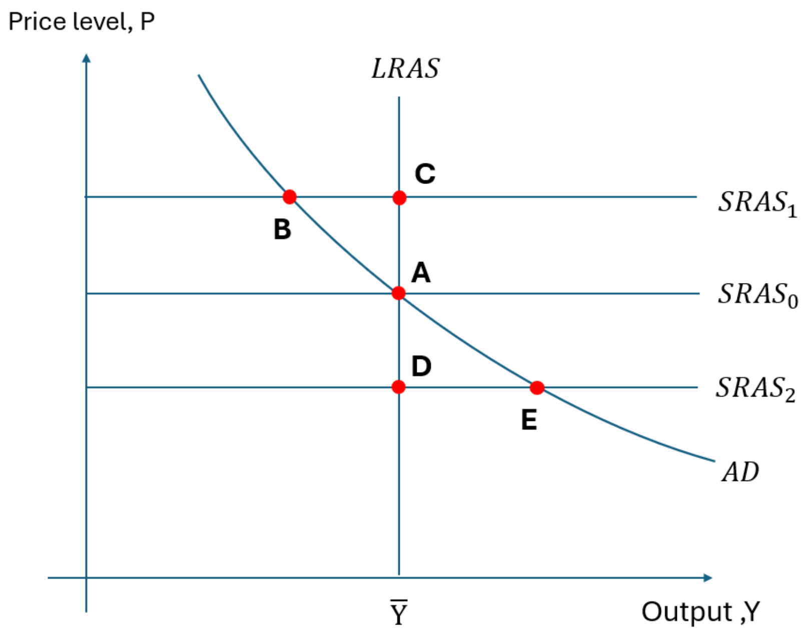
7. Suppose that the government wants to raise investment but keep output constant. In the IS–LM model, what mix of monetary and fiscal policy will achieve this goal?

   (a) Raising government spending and increasing money supply.
   (b) Raising government spending and decreasing money supply.
   (c) Lowering government spending and increasing money supply.
   (d) Lowering government spending and decreasing money supply.

8. Stabilization policy refers to policy actions aimed at:

   (a) reducing the severity of short-run economic fluctuations.
   (b) equalizing incomes of households in the economy.
   (c) maintaining constant shares of output going to labor and capital.
   (d) preventing increases in the poverty rate.

9. Assume that the economy is at point E. With no further shocks or policy moves, the economy in the long run will be at point:



   (a) A.
   (b) B.
   (c) C.
   (d) D.
   (e) E.

10. Consider an economy described by the production function

$$Y_t = 0.5\sqrt{K_t}\sqrt{L_t}.$$

The economy has a saving rate of 20 percent, a depreciation rate of 5 percent. $L_t = 1$ for all periods (no population growth), and there is no technological progress. The long-run growth rate of output per worker is ________ %.

(a) 0
(b) 1
(c) 5
(d) 20
(e) 22

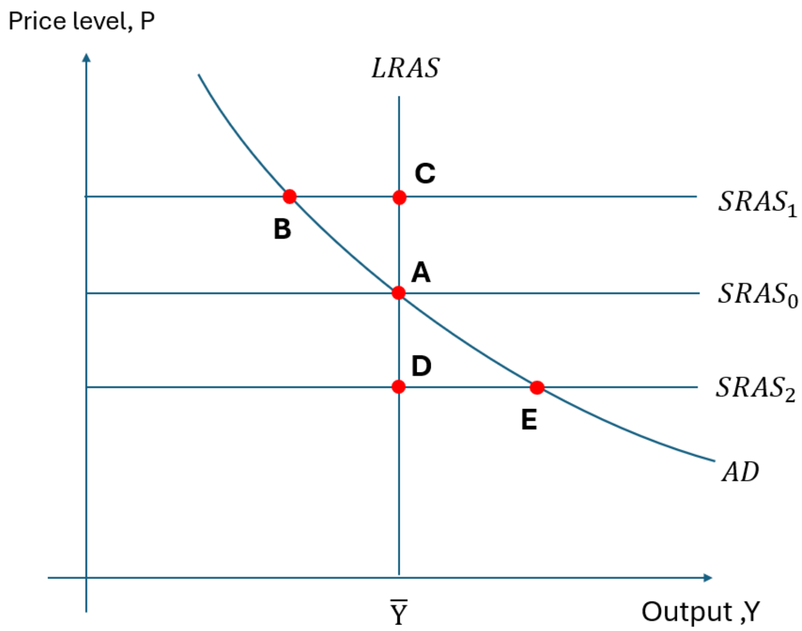
11. Consider an endogenous growth model with the production function

$$Y_t = 0.5K_tL_t.$$

The economy has a saving rate of 20 percent, a depreciation rate of 5 percent. $L_t = 1$ for all periods (no population growth), and there is no technological progress. The long-run growth rate of output per worker is ________ %. (*Hint: First, use capital per worker evolution equation to determine the growth rate of k. Then proceed with the growth rate of y.*)

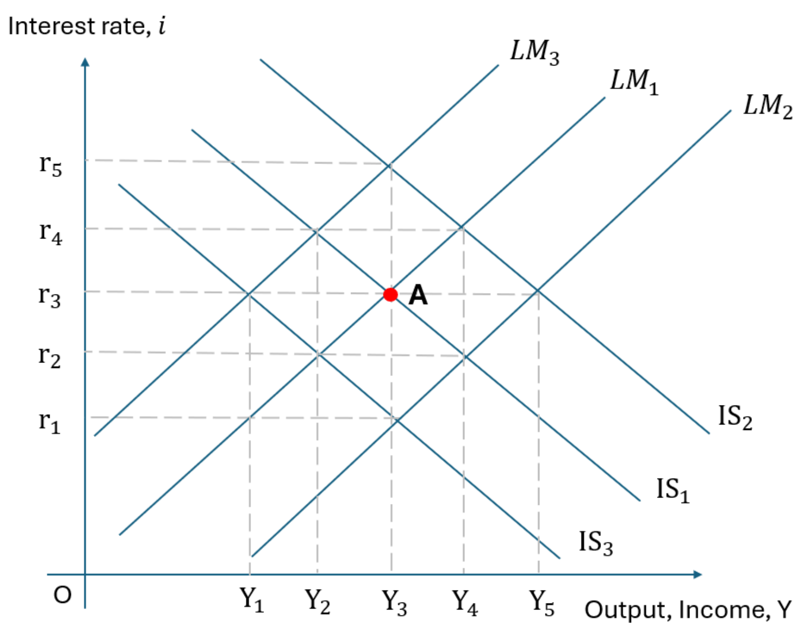
(a) 0
(b) 3
(c) 5
(d) 7
(e) 12

12. Assume that the economy starts at point A, and there is a drought that severely reduces agricultural output in the economy for just one year. In this situation, point ________ represents the short-run equilibrium immediately following the drought, and point ________ represents the eventual long-run equilibrium.



(a) B; C
(b) B; A
(c) E; D
(d) C; A
(e) E; A

13. Based on the graph below, starting from equilibrium at point A (combination of interest rate r_3 and income Y_3), a decrease in taxes would generate the new equilibrium combination of interest rate and income:



(a) r_5, Y_3

(b) r_2, Y_2

(c) r_4, Y_4

(d) r_3, Y_5

(e) r_1, Y_3

14. An economy begins in long-run equilibrium. A negative supply shock raises firms' costs temporarily. At the same time, the government increases spending, and the central bank keeps the nominal money supply fixed. Which outcome is most likely in the short run and long run?

    (a) Short run: output falls and inflation rises; long run: output stays below natural output and inflation keeps rising.

    (b) Short run: output rises and inflation falls; long run: output returns to natural output with a permanently lower price level.

    (c) Short run: the fiscal expansion partially offsets the supply shock, so output falls less than it otherwise would and inflation rises more than it otherwise would; long run: output returns to natural output, but the price level is higher than initially.

    (d) Short run: output is unchanged because fiscal policy and the supply shock exactly cancel; long run: output and the price level both return to their initial levels.

15. Assume the economy starts from long-run equilibrium in the AD–AS framework with sticky prices in the short run. If the Federal Reserve increases the money supply, then ____________ increase(s) in the short run, and ____________ increase(s) in the long run.

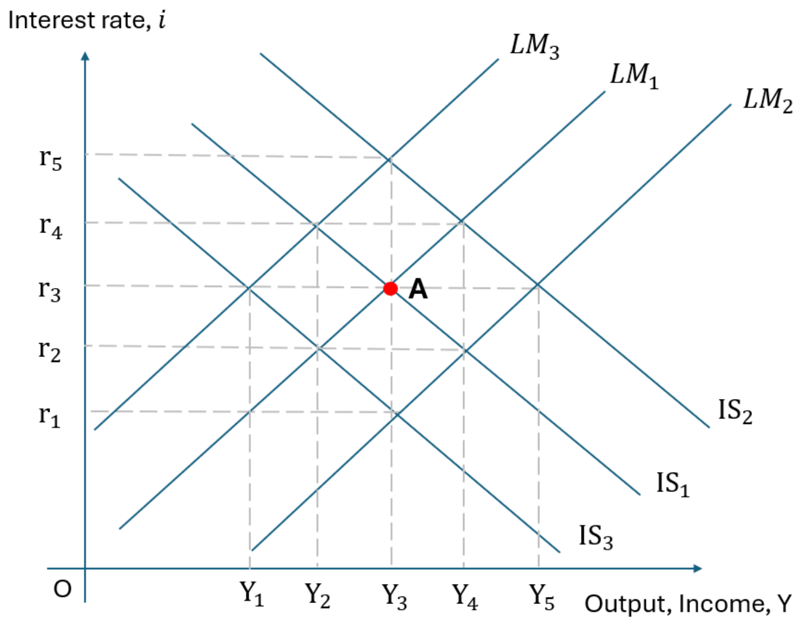
    (a) output; prices

    (b) output; output

    (c) prices; output

    (d) prices; prices

16. Based on the graph, starting from equilibrium at point A (combination of interest rate r_3 and income Y_3), an increase in the money supply would generate the new equilibrium combination of interest rate and income:



(a) r_5, Y_3

(b) r_2, Y_4

(c) r_3, Y_5

(d) r_5, Y_5

(e) r_4, Y_2

17. If an economy is in a steady state with no population growth or technological change and the marginal product of capital is less than the depreciation rate:

(a) the economy is following the Golden Rule.

(b) steady-state consumption per worker would be higher in a steady state with a lower saving rate.

(c) steady-state consumption per worker would be higher in a steady state with a higher saving rate.

(d) the depreciation rate should be decreased to achieve the Golden Rule level of consumption per worker.

Answer the following two questions. Consider an economy with the following production function:

$$Y_t = A_t K_t^{\alpha} L_t^{1-\alpha},$$

where Y_t denotes output, A_t is an efficiency factor capturing technological progress, K_t denotes physical capital, and L_t denotes labor. This economy has an aggregate wage bill of 70 billion dollars and an aggregate GDP of 100 billion.

The following table presents the growth rate of the US GDP, capital stock, and labor for 1948–1973, and 1973–2016. Find the technology growth rate for each period using the following equation:

$$g_A = g_Y - \alpha g_K - (1 - \alpha) g_L.$$

	Sources of Growth (average percentage increase per year)			
Years	**Output Growth**	**Capital Growth**	**Labor Growth**	**Total Factor Productivity**
1948–1973	4.3	4.3	1.4	?
1973–2016	3	4	1.4	?

18. The growth rate of TFP between 1948–1973 was __________ %. The growth rate of TFP between 1973–2016 was __________ %. Please round your solutions to the first decimal place, for example, 0.17 to 0.2. (*Hint: recall what α corresponds to for a given production function.*)

    (a) 2.0, 0.8
    (b) 1.7, 2.3
    (c) 2.6, 1.4
    (d) 1.4, 2.6
    (e) 2.3, 1.7

19. Based on your the answer in the previous question and information from the table, we observe a slowdown in output growth due to the following factor(s):

    (a) A slowdown in TFP growth
    (b) A slowdown in capital growth
    (c) A slowdown in TFP growth and capital growth
    (d) A slowdown in labor growth and capital growth
    (e) A slowdown in all three factors

20. The predicted effects of $1 billion increase in government spending on output is __________ in the Keynesian cross model than the IS-LM model, because the IS-LM model also accounts for the equilibrium effects of changes in __________.

    (a) larger; the interest rate
    (b) smaller; taxes
    (c) larger; taxes
    (d) smaller; the interest rate

21. Two identical countries, Country A and Country B, can be described by the IS-LM model in the short run. The governments of both countries increased government spending by the same amount to stimulate the economy. The Central Bank of A follows a policy of holding interest rate constant and the Central Bank of B follows a policy of keeping money supply constant. Output rises:

    (a) more in Country A.
    (b) more in Country B.
    (c) the same in both countries.
    (d) indeterminately in the two countries.

22. In the Keynesian cross model, suppose the marginal propensity to consume is 0.8. If the government cuts taxes by \$100, equilibrium income will:

    (a) increase by \$500
    (b) increase by \$400
    (c) increase by \$100
    (d) increase by \$80
    (e) not change

23. If the government increases both spending and taxes by the same amount (\$1 billion), in the Keynesian cross model, equilibrium income will:

    (a) not change
    (b) increase by \$1 billion
    (c) increase by more than \$1 billion
    (d) increase by less than \$1 billion

24. Consider the IS-LM model in the short run and the AD-AS model in the long run. A permanent increase in taxes:

    (a) lowers income and the interest rate in the short run, but in the long run income returns to its natural level, consumption falls, and investment rises.
    (b) lowers income and the interest rate in the short run, but in the long run income returns to its natural level, consumption rises, and investment falls.
    (c) lowers income and the interest rate in both the short run and the long run.
    (d) not enough data to determine

25. When the Federal Reserve reduces the money supply, at a given price level the amount of output demanded is __________, and the aggregate demand curve shifts to the __________.

    (a) greater; left

(b) greater; right

(c) lower; left

(d) lower; right

26. Starting from long-run equilibrium, if a lockdown order pushes up auto prices throughout the economy, the Fed could move the economy more rapidly back to full employment output by:

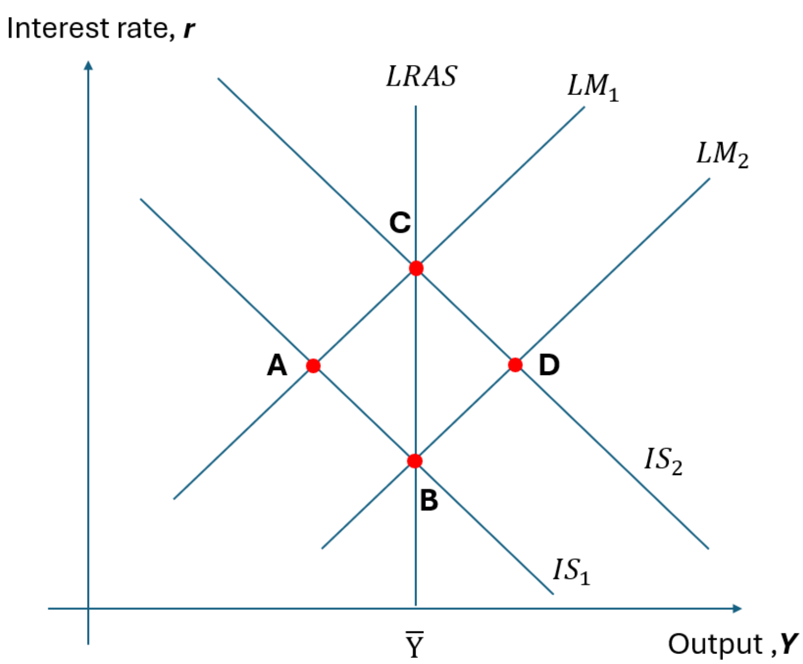
   (a) increasing the money supply, but at the cost of permanently higher prices.

   (b) decreasing the money supply, but at the cost of permanently lower prices.

   (c) increasing the money supply, which would restore the original price level.

   (d) decreasing the money supply, which would restore the original price level.

27. Based on the graph, if the economy starts from a short-run equilibrium at A, then the long-run equilibrium will be at __________, with a __________ price level.



   (a) C; lower

   (b) B; lower

   (c) C; higher

   (d) B; higher

28. The IS curve either shifts or rotates when any of the following economic variables change except:

   (a) the interest rate.

(b) government spending.

(c) taxes.

(d) the marginal propensity to consume.

29. Two identical countries, Country A and Country B, can each be described by a Keynesian-cross model. The MPC is 0.8 in Country A and MPC is 0.9 in Country B. Country A decides to increase spending by $2 billion, while Country B decides to cut taxes by $2 billion. In which country will the new equilibrium level of income be greater?

    (a) Country A

    (b) Country B

    (c) The same in both countries

    (d) Unknown

30. An increase in government spending raises income:

    (a) and the interest rate in the short run but leaves both unchanged in the long run.

    (b) in the short run but leaves it unchanged in the long run, while lowering investment.

    (c) in the short run but leaves it unchanged in the long run, while lowering consumption.

    (d) and the interest rate in both the short run and in the long run.

31. Policymakers are contemplating undertaking either an increase in government spending or an increase in the money supply. Either policy is forecast to have the same impact on income in the short run. Use the IS-LM model to compare the impact on consumption and investment of the two policy alternatives.

    (a) Both policies have different effects on both investment and consumption.

    (b) Both policies have the same effect on investment and consumption.

    (c) Both policies have the same effect on investment but not consumption.

    (d) Both policies have the same effect on consumption but not investment.

Consider an economy described by the following IS-LM model:

$$C = 100 + 0.75(Y - T)$$

$$I = 200 - 25r$$

$$G = 100, \quad T = 100$$

$$\frac{M}{P} = Y - 100r$$

$$M = 1000, \quad P = 2$$

Answer the following 3 questions:

32. The equilibrium interest rate is ________ and the equilibrium output is ________.

    (a) r=4, Y=900
    (b) r=6, Y=1100
    (c) r=2, Y=700
    (d) r=8, Y=1300

33. Now suppose the government increases spending from G=100 to G=150. Output will then be equal to 1000 in equilibrium. Then, the government spending multiplier is: (*Hint: note, the government spending multiplier in IS-LM model is different from* $\frac{1}{1-MPC}$ *that we have in Keynesian Cross model, because of "crowding out" effect. Therefore, you cannot use that formula.*)

    (a) 1.5
    (b) 2
    (c) 2.5
    (d) 4

34. What is the absolute value of "crowding out" effect for private investment from an increase in G from 100 to 150, based on the model? (*Hint: crowding out refers to the reduction in private investment caused by an increase in government spending. Using your answer from (32), compute investment at* $G = 100$*; then find the new equilibrium at* $G = 150$ *and compute investment again.*)

    (a) 25
    (b) 50
    (c) 100
    (d) 150

Use this page for your calculations: